

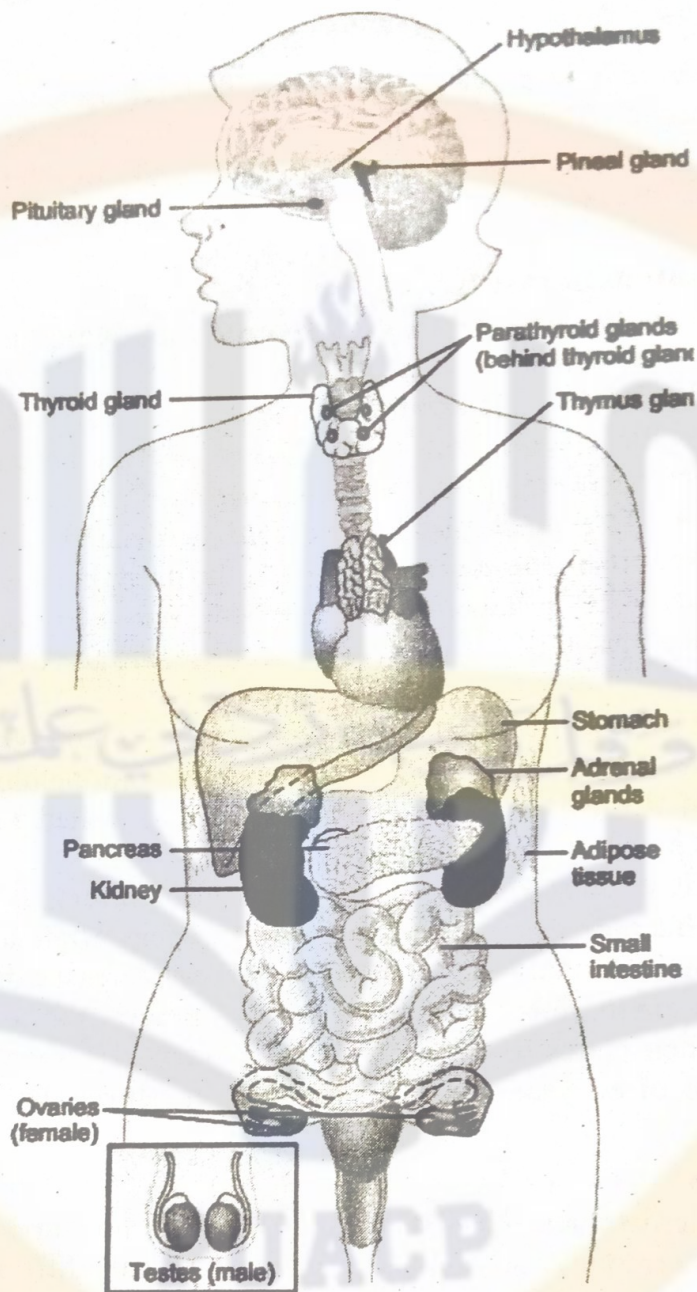


Fig: Different Endocrine Glands in the Body

HYPOTHALAMUS

Almost all secretion by the pituitary is controlled by hormonal or nervous signals from the hypothalamus. In an experiment, pituitary gland was removed from its normal position beneath the hypothalamus. It was transplanted to some other part of the body. As a result, the rates of secretion of the different hormones (except for prolactin) fell to very low levels.

- a) Secretion from the posterior pituitary is controlled by nerve signals originate in hypothalamus. These terminate in the posterior pituitary.
- b) The secretions by the anterior pituitary are controlled by hormones called **hypothalamic releasing and hypothalamic inhibitory hormones** (or factors). These are secreted within the hypothalamus itself. Then it is conducted to the anterior pituitary through minute blood vessels called **hypothalamic-hypophyseal portal vessels**. In the anterior pituitary, these releasing and inhibitory hormones act on glandular cells to control their secretion.

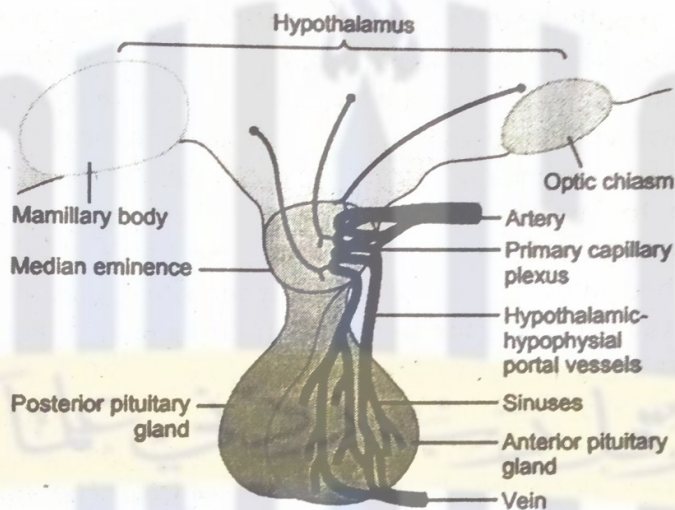


Fig: Hypothalamic-hypophyseal portal system

The hypothalamus receives signals from many sources in the nervous system.

1. When a person is exposed to pain, a portion of the pain signal is transmitted into the hypothalamus.
2. Similarly, when a person experiences some powerful depressing or exciting thought, a portion of the signal is transmitted into the hypothalamus.
3. Olfactory stimuli denoting pleasant or unpleasant smells transmit strong signal components directly and through amygdaloid nuclei into the hypothalamus.
4. Even the concentrations of nutrients, electrolytes, water, and various hormones in the blood excite or inhibit various portions of the hypothalamus.

Thus, the hypothalamus is a collecting center for information coming from the body. Much of this information is used to control secretions of the many globally important pituitary hormones.

Hypothalamic-Hypophyseal Portal Blood Vessels of Anterior Pituitary Gland

The anterior pituitary is a highly vascular gland. There are extensive capillary sinuses among the glandular cells. Almost all the blood that enters these sinuses passes first through another capillary bed in the lower hypothalamus. The blood then flows through small hypothalamic-hypophyseal portal blood vessels into the anterior pituitary sinuses.

Median eminence

Lowermost portion of the hypothalamus is called median eminence. It connects inferiorly with the pituitary stalk. Small arteries penetrate into the substance of the median eminence. Then additional small vessels return to its surface, coalescing to form the hypothalamic-hypophyseal portal blood vessels. These pass downward along the pituitary stalk to supply blood to anterior pituitary sinuses.

Hypothalamic Releasing and Inhibitory Hormones Are Secreted into Median Eminence

Special neurons in the hypothalamus synthesize and secrete the hypothalamic releasing and inhibitory hormones. These hormones control secretion of the anterior pituitary hormones. These neurons originate in various parts of the hypothalamus. These send their nerve fibers to the median eminence and tuber cinereum. Tuber cinereum is an extension of hypothalamic tissue into the pituitary stalk. The endings of these fibers are different from most endings in the central nervous system. Their function is to secrete the hypothalamic releasing and inhibitory hormones into the tissue fluids. These hormones are immediately absorbed into the hypothalamic-hypophyseal portal system. These are then carried directly to the sinuses of the anterior pituitary gland.

Hypothalamic Releasing and Inhibitory Hormones

The function of the releasing and inhibitory hormones is to control secretion of the anterior pituitary hormones. Mostly releasing hormones are important. But a **prolactin** is a hypothalamic inhibitory hormone. It exerts more control. The major hypothalamic releasing and inhibitory hormones are following:

1. **Thyrotropin-releasing hormone (TRH):** It causes release of thyroid-stimulating hormone
2. **Corticotropin-releasing hormone (CRH):** It causes release of adrenocorticotropin
3. **Growth hormone-releasing hormone (GHRH):** It causes release of growth hormone
4. **Growth hormone inhibitory hormone (GHIH):** It is also called **somatostatin**. It inhibits release of growth hormone
5. **Gonadotropin-releasing hormone (GnRH):** It causes release of the two gonadotropic hormones, luteinizing hormone and follicle-stimulating hormone
6. **Prolactin inhibitory hormone (PIH):** It causes inhibition of prolactin secretion

There are some additional hypothalamic hormones. These include one that stimulates prolactin secretion. The others are that which inhibit release of the anterior pituitary hormones.

Specific Areas in Hypothalamus Control Secretion of Specific Hypothalamic Releasing and Inhibitory Hormones

All or most of the hypothalamic hormones are secreted at nerve endings in the median eminence. Then these are transported to the anterior pituitary gland. Electrical stimulation of this region excites these nerve endings. Therefore, this causes release of essentially all hypothalamic hormones. Different neuronal cell bodies give rise to median eminence

nerve endings. These neuronal cells bodies are located in other discrete areas of the hypothalamus or in closely related areas of the basal brain. The specific loci of the neuronal cell bodies forming different hypothalamic releasing or inhibitory hormones are still poorly known.

Table

Hypothalamic Releasing and Inhibitory Hormones That Control Secretion of the Anterior Pituitary Gland

Hormone	Structure	Primary Action on Anterior Pituitary
Thyrotropin-releasing hormone (TRH)	Peptide of 3 amino acids	Stimulates secretion of TSH by thyrotropes
Gonadotropin-releasing hormone (GnRH)	Single chain of 10 amino acids	Stimulates secretion of FSH and LH by gonadotropes
Corticotropin-releasing hormone (CRH)	Single chain of 41 amino acids	Stimulates secretion of ACTH by corticotropes
Growth hormone-releasing hormone (GHRH)	Single chain of 44 amino acids	Stimulates secretion of growth hormone by somatotropes
Growth hormone inhibitory hormone (somatostatin)	Single chain of 14 amino acids	Inhibits secretion of growth hormone by somatotropes
Prolactin-inhibiting hormone (PIH)	Dopamine (a catecholamine)	Inhibits secretion of prolactin by lactotropes

PITUITARY GLAND

The pituitary gland is also called the **hypophysis**. It is a small gland about 1 centimeter in diameter and 0.5 to 1 gram in weight. It lies in the **sella turcica**. It is a bony cavity at the base of the brain. Pituitary gland is connected to hypothalamus by pituitary (or hypophyseal) stalk. Physiologically, pituitary gland is divisible into two distinct portions:

- Anterior pituitary:** It also called adenohypophysis.
- Posterior pituitary:** It is also called neurohypophysis.
- Pars intermedia:** It is present between anterior and posterior parts of pituitary gland. It is a small avascular zone. It is almost absent in the human being. But it is much larger and much more functional in some lower animals. It is mostly a glandular tissue. It secretes **melanophore-stimulating hormone** in response to external stimuli. This hormone induces changes in the coloration of the body surface of many animals.

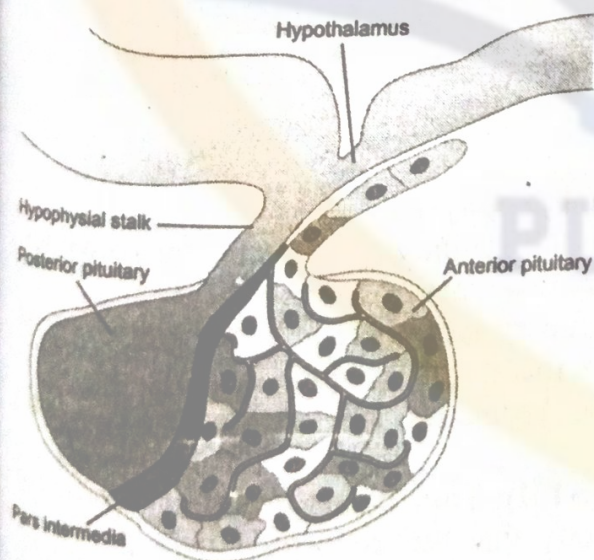


Fig: Pituitary gland

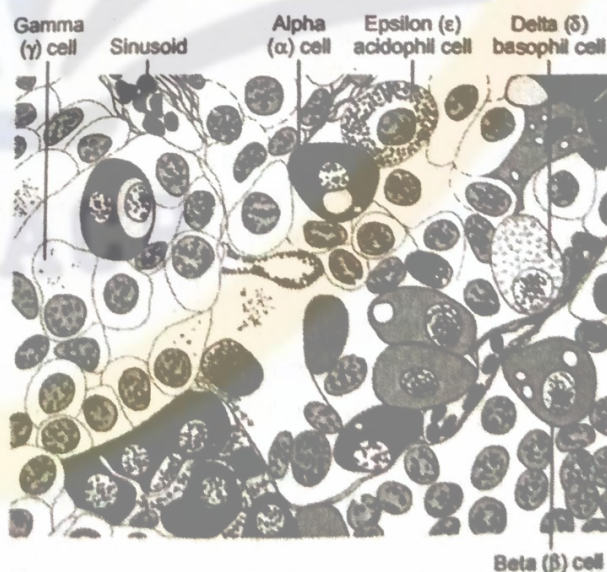


Fig: Cellular structure of pituitary gland

Origin of Pituitary Gland

In embryo, two portions of the pituitary originate from different sources

- Anterior pituitary arises from Rathke's pouch. It is an embryonic invagination of pharyngeal epithelium. The origin of anterior pituitary from pharyngeal epithelium explains epithelioid nature of its cells.
- Posterior pituitary from a neural tissue outgrowth from the hypothalamus. The origin of posterior pituitary from neural tissue explains presence of large numbers of glial-type cells in this gland.

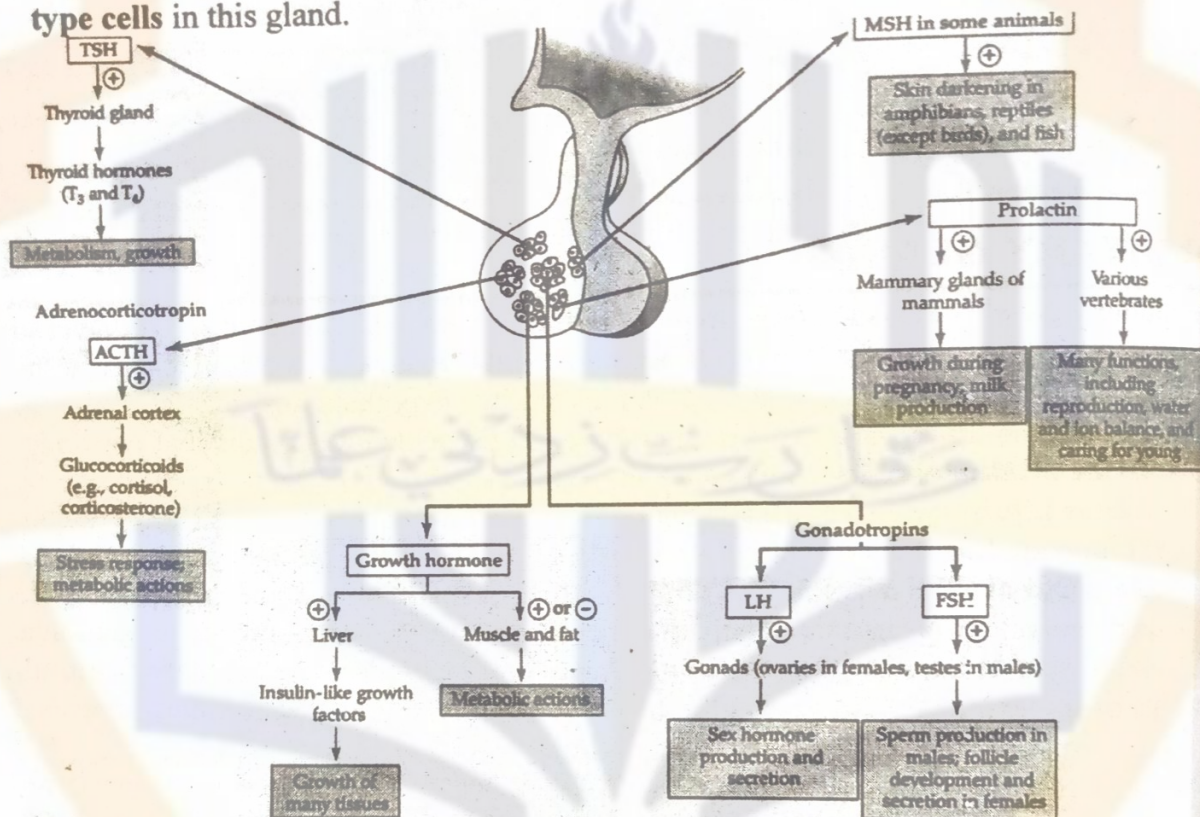


Fig: Metabolic functions of the anterior pituitary hormones

Hormone of Pituitary Gland

(a) Hormones of Anterior Pituitary (Adenohypophysis)

It is the true endocrine part of the pituitary gland. It synthesizes six different hormones. All of these hormones are polypeptides. There are two types of hormones:

- Tropic hormones:** The hormones whose primary target is another endocrine gland are called tropic hormone. Four hormones are true tropic hormones.
- Nontropic hormones:** The two nontropic hormones are growth hormone and prolactin.

- Growth hormone (GH) or somatotropin (STH):** They do not influence a particular target tissue. It affects all parts of the body that are concerned with growth. It stimulates the uptake of amino acids, RNA synthesis, and ribosome activity. Thus it directly induces the cell division and protein synthesis in most types of cells.

2. **Prolactin (PRL):** It has the widest range of actions. It plays an important role in reproduction.
 - It stimulates reproductive migrations in many mammals like elk and caribou.
 - Prolactin also enhances mammary gland development and milk production in female mammals.
3. **Thyrotropin or thyroid - stimulating hormone (TSH):** It stimulates the development of thyroid gland and its secretion of thyroxine. Thyroxine is the main thyroid hormone.
4. **Adrenocorticotropic hormone (ACTH):** It stimulates the adrenal gland. Adrenal gland secretes steroid hormones called **glucocorticoids (cortisol)**. Corticotropin-releasing factor from the hypothalamus regulates the secretion of ACTH. The secretion of CRF itself is regulated by a feedback system. This system is stimulated by factors like stress, insulin, ADH and other hormones.

Gonadotropins hormones of adenohypophysis

Adenohypophysis secretes two gonadotropins hormones:

1. **Luteinizing hormone:** It has different roles in female and males:
 - (a) **Female:** Luteinizing hormone (LH) receives its name from the corpus luteum. Corpus luteum is a temporary endocrine tissue in the ovaries. It secretes the female sex hormones estrogen and progesterone. The increases level of LH in blood stimulates ovulation. **The release of a mature egg from an ovary is called ovulation.**
 - (b) **Male:** LH affects the testes. Testes secrete the male hormone testosterone.
2. **Follicle-stimulating hormone (FSH):**
 - (a) **Female:** It stimulates the follicular cells in the ovaries. They develop in to mature eggs. Ovaries also produce estrogen.
 - (b) **Males:** FSH stimulates the cells of the testes to produce sperm in males.

Hormones of Posterior Pituitary (Neurohypophysis)

Two important peptide hormones are secreted by the posterior pituitary.

1. **Antidiuretic hormone:** Diuretics stimulate urine excretion. But antidiuretic decrease urine secretion. Sometimes, mammal loses water and becomes dehydrated. Now antidiuretic hormone (ADH or vasopressin) is released. It increases water absorption in the kidneys. Thus less urine is secreted. Therefore, kidney retains water. This negative feedback system thus restores water and solute homeostasis.
2. **Oxytocin:** Oxytocin plays a role in mammalian reproduction. It effects on smooth muscle. It stimulates contraction of the uterus or uteri and help in expulsion of the offspring during birth. It promotes the ejection of milk from the mammary glands. It provides nourishment to the newborn.

Evolutionary origin of ADH and Oxytocin

It is believed that both ADH and oxytocin are evolved from the same ancestral messenger. This messenger control water and solute concentration. For example, neurohypophysis is larger in animals that live in arid parts of the world. They need to

conserve water. Both these hormones have similar structures. There is only one difference of two amino acids.

Posterior Pituitary Hormones Synthesized by Cell Bodies in Hypothalamus

The bodies of the cells that secrete the posterior pituitary hormones are not located in the pituitary gland itself. These are large neurons called **magnocellular neurons**. These are located in the supraoptic and paraventricular nuclei of the hypothalamus. The hormones are then transported in the axoplasm of neurons' nerve fibers. These nerve fibers pass from the hypothalamus to posterior pituitary gland.

Table:

Cells and Hormones of the Anterior Pituitary Gland and Their Physiological Functions

Cell	Hormone	Chemistry	Physiological Actions
Somatotropes	Growth hormone (GH; somatotropin)	Single chain of 191 amino acids	Stimulates body growth; stimulates secretion of IGF-1; stimulates lipolysis; inhibits actions of insulin on carbohydrate and lipid metabolism
Corticotropes	Adrenocorticotrophic hormone (ACTH; corticotropin)	Single chain of 39 amino acids	Stimulates production of glucocorticoids and androgens by the adrenal cortex; maintains size of zona fasciculata and zona reticularis of cortex
Thyrotropes	Thyroid-stimulating hormone (TSH; thyrotropin)	Glycoprotein of two subunits, α (89 amino acids) and β (112 amino acids)	Stimulates production of thyroid hormones by thyroid follicular cells; maintains size of follicular cells
Gonadotropes	Follicle-stimulating hormone (FSH)	Glycoprotein of two subunits, α (89 amino acids) and β (112 amino acids)	Stimulates development of ovarian follicles; regulates spermatogenesis in the testis
	Luteinizing hormone (LH)	Glycoprotein of two subunits, α (89 amino acids) and β (115 amino acids)	Causes ovulation and formation of the corpus luteum in the ovary; stimulates production of estrogen and progesterone by the ovary; stimulates testosterone production by the testis
Lactotropes, Mammatropes IGF, insulin-like growth factor	Prolactin (PRL)	Single chain of 198 amino acids	Stimulates milk secretion and production

Cells Types of Anterior Pituitary Gland

There is one cell type for each major hormone formed in the anterior pituitary gland. At least five cell types can be differentiated in the anterior pituitary gland. These five cell types are:

1. Somatotropes or growth hormone (hGH)
2. Corticotropes—adrenocorticotropin (ACTH)
3. Thyrotropes—thyroid-stimulating hormone (TSH)
4. Gonadotropes—gonadotropic hormones. It includes both luteinizing hormone (LH) and follicle stimulating hormone (FSH)
5. Lactotropes—prolactin (PRL)

About 30 to 40 per cent of the anterior pituitary cells are Somatotropes. They secrete growth hormone. About 20 per cent are corticotropes that secrete ACTH. The each remaining forms 3 to 5 per cent. Somatotropes stain strongly with acid dyes and are therefore called **acidophils**.

THYROID GLAND

The thyroid gland is located immediately below the larynx on each side of and anterior to the trachea. It is one of the largest of the endocrine glands. It normally weighs 15 to 20 grams in adults.

Structure of Thyroid Gland

Thyroid gland is composed of large numbers of closed **follicles** (100 to 300 micrometers in diameter). These follicles are filled with a secretory substance called **colloid**. These are lined with cuboidal epithelial cells. These cells secrete into the interior of the follicles. The major constituent of colloid is large glycoprotein **thyroglobulin**. Thyroglobulin contains thyroid hormones within its molecule. The secretion enters the follicles. Then it must be absorbed back through follicular epithelium into the blood. Now it can function in the body. The thyroid gland has a blood flow about five times the weight of the gland each minute.

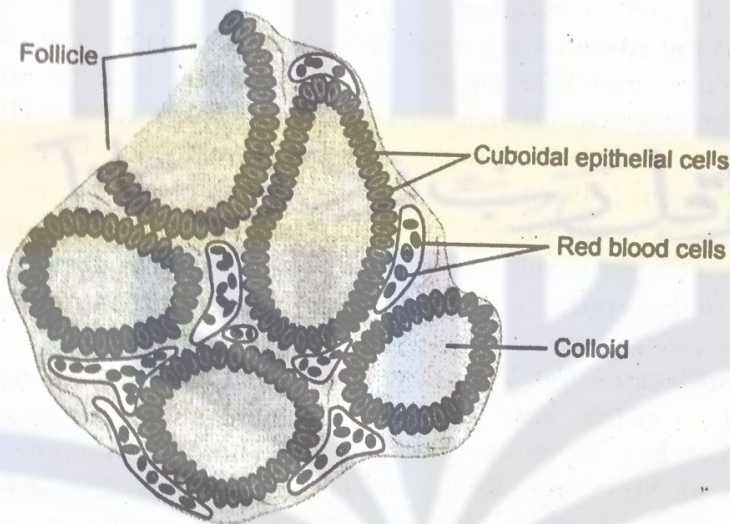


Fig: Structure of Thyroid Gland

Hormones Secreted by Thyroid Gland

Thyroid gland secretes following hormones:

- Thyroxine and triiodothyronine:** Thyroid secretes two major hormones, thyroxine and triiodothyronine. These are commonly called T₄ and T₃, respectively.
- Calcitonin:** The thyroid gland also secretes **calcitonin**. It is an important hormone for calcium metabolism.

Thyroxine (T₃) and Triiodothyronine (T₄)

Thyroid secretes two major hormones, thyroxine and triiodothyronine. Both of these hormones profoundly increase the metabolic rate of the body. Complete lack of thyroid secretion causes the basal metabolic rate to fall 40 to 50 per cent below normal. The extreme excesses of thyroid secretion can increase the basal metabolic rate to 60 to 100

per cent above normal. Thyroid secretion is controlled primarily by thyroid-stimulating hormone (TSH). It is secreted by the anterior pituitary gland.

About 93 per cent of the metabolically active hormones secreted by thyroid gland is thyroxine, and 7 per cent triiodothyronine. But almost all the thyroxine is converted to triiodothyronine in the tissues. The functions of these two hormones are qualitatively the same. But they differ in rapidity and intensity of action. Triiodothyronine is about four times as potent as thyroxine. But it is present in the blood in much smaller quantities and persists for a much shorter time than thyroxine.

Physiological Roles of Thyroxine and Triiodothyronine

1. **Stimulation of Carbohydrate Metabolism:** Thyroid hormone stimulates almost all aspects of carbohydrate metabolism. They cause rapid uptake of glucose by the cells, enhanced glycolysis, enhanced gluconeogenesis, increased rate of absorption from the gastrointestinal tract, and even increased insulin secretion with its resultant secondary effects on carbohydrate metabolism.
2. **Stimulation of Fat Metabolism.** They also enhance all aspects of fat metabolism. In particular, lipids are mobilized rapidly from the fat tissue. It decreases the fat stores of the body to a greater extent than almost any other tissue element. This also increases the free fatty acid concentration in the plasma and greatly accelerates the oxidation of free fatty acids by the cells.
3. **Effect on Plasma and Liver Fats:** Increased thyroid hormone decreases the concentrations of cholesterol, phospholipids, and triglycerides in the plasma. The decreased thyroid secretion greatly increases plasma concentrations of cholesterol, phospholipids, and triglycerides.
4. **Increased Requirement for Vitamins:** The thyroid hormone increases the quantities of many bodily enzymes. The vitamins are essential parts of some of the enzymes or coenzymes. Therefore, thyroid hormone increased need for vitamins.
5. **Increased Basal Metabolic Rate:** The thyroid hormone increases metabolism in almost all cells of the body. Therefore, excessive quantities of the hormone can increase the basal metabolic rate 60 to 100 per cent above normal. When no thyroid hormone is produced, then the basal metabolic rate falls almost to one-half normal. Extreme amounts of the hormones are required to cause very high basal metabolic rates.
6. **Decreased Body Weight:** Greatly increased thyroid hormone decreases the body weight. The decreased hormone increases the body weight. Thyroid hormone also increases the appetite. Therefore, these effects do not always occur.
7. **Effect of Thyroid Hormones on the Cardiovascular System:** Increased metabolism in tissues causes more rapid utilization of oxygen. Similarly, release of metabolic end products from the tissues is also increased. These effects cause vasodilation in most body tissues. Thus it increases blood flow. As a result, due to increased blood flow, cardiac output also increases,

8. **Increased Heart Rate:** The heart rate increases considerably more under the influence of thyroid hormone. Therefore, thyroid hormone has a direct effect on the excitability of the heart. It increases the heart rate.
9. **Increased Heart Strength:** The increased enzymatic activity increases the strength of the heart.
10. **Arterial Pressure:** The mean arterial pressure remains about normal after taking thyroid hormone. The increased blood flow through tissues between heartbeats also increases the pulse pressure. Hence the systolic pressure in hyperthyroidism is increased by 10 to 15 mm Hg
11. **Increased Respiration:** The increased rate of metabolism increases the utilization of oxygen and formation of carbon dioxide. These effects activate all the mechanisms that increase the rate and depth of respiration.
12. **Increased Gastrointestinal Motility:** Thyroid hormone increases both rates of secretion of digestive juices and motility of gastrointestinal tract.
13. **Excitatory Effects on the Central Nervous System:** In general, thyroid hormone increases the rapidity of cerebation
14. **Effect on the Function of the Muscle:** Slight increase in thyroid hormone makes the muscles react with vigor. But when the quantity of hormone becomes excessive, the muscles become weakened because of excess protein catabolism.
15. **Effect on Other Endocrine Glands:** Increased thyroid hormone increases the rates of secretion of most other endocrine glands.

Regulation of Thyroid Hormone Secretion

Right amount of thyroid hormone must be secreted at all times to maintain normal levels of metabolic activity in the body. For this purpose, specific feedback mechanisms operate through the hypothalamus and anterior pituitary gland to control the rate of thyroid secretion. These mechanisms are as follows.

(a) TSH (from the Anterior Pituitary Gland) Increases Thyroid Secretion

TSH increases the secretion of thyroxine and triiodothyronine by the thyroid gland. Its specific effects on the thyroid gland are as follows:

1. It increases proteolysis of thyroglobulin that has already been stored in the follicles. It releases thyroid hormones into the circulating blood
2. It increases activity of iodide pump. This pump increases the rate of "iodide trapping" in the glandular cells,
3. Increased iodination of tyrosine to form the thyroid hormones
4. Increased size and increased secretory activity of the thyroid cells
5. Increased number of thyroid cells plus a change from cuboidal to columnar cells

(b) Anterior Pituitary Secretion of TSH Is Regulated by Thyrotropin-Releasing Hormone from the Hypothalamus

Anterior pituitary secretion of TSH is controlled by a hypothalamic hormone, thyrotropin-releasing hormone (TRH). It is secreted by nerve endings in the median eminence of hypothalamus. From the median eminence, the TRH is then transported to

2. It also increases the calcium absorption from the small intestine.
3. It also promotes calcium reabsorption in the kidney tubules. It decreases the amount of calcium excreted in the urine.

Endocrine Control of Calcium Metabolism

Calcium is regulated in both the extracellular and intracellular fluids. The intracellular Ca^{2+} concentration is kept very low ($<10^{-7}$ M). The extracellular concentration of Ca^{2+} is higher than in cytoplasm. But it too is maintained within very narrow limits.

- a) Low levels of Ca^{2+} cause hyperexcitability of nerve and muscle membranes. It can produce muscle twitches and sensations of tingling and numbness.
- b) High levels of Ca^{2+} cause reduced excitability that produces hyporeflexia (slower reflexes) and lethargy.

The total amount of calcium in the blood plasma includes that:

- a) Bound to proteins (mainly albumin)
- b) Complexed with anions like phosphate, sulfate, and citrate,
- c) Free as ionized Ca^{2+} . Only free Ca^{2+} (about half the total calcium) is biologically active.

(a) Response to low Ca^{2+}

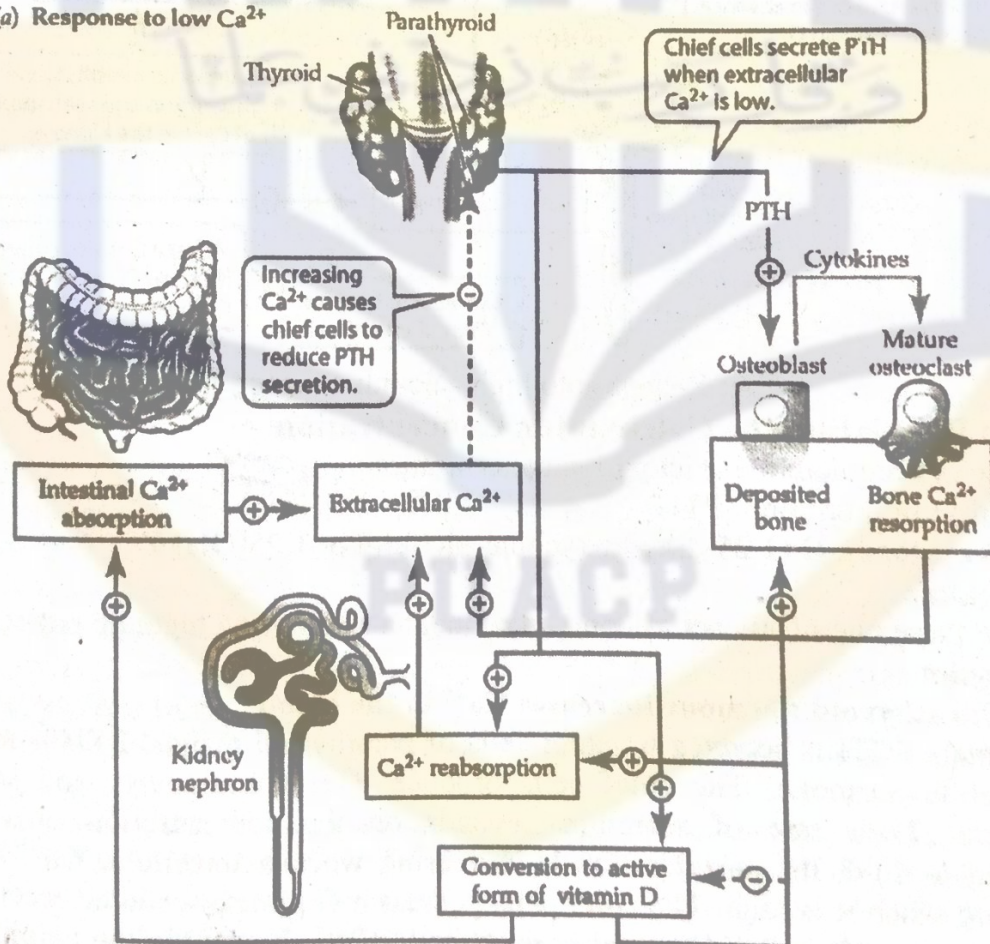


Fig: Response to low calcium concentration

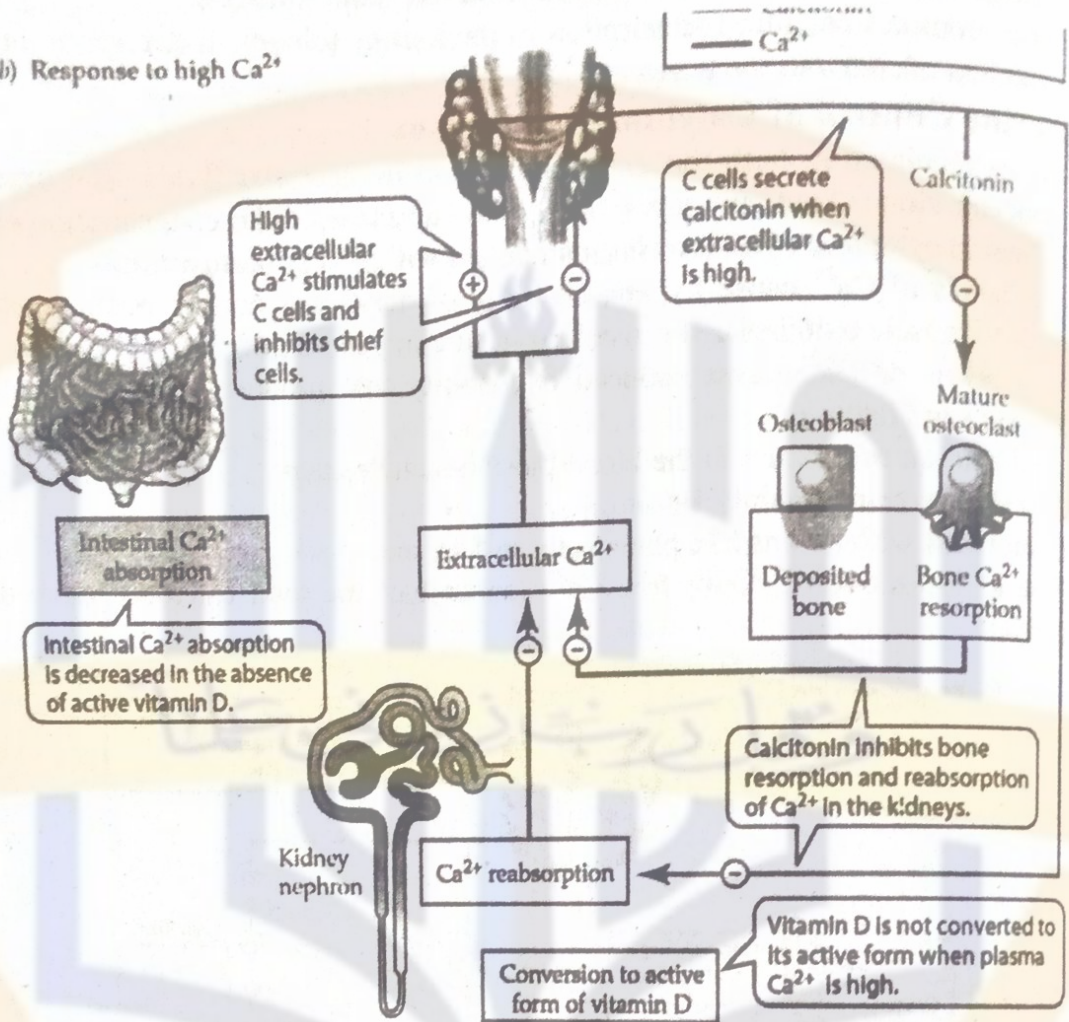
(b) Response to high Ca^{2+} 

Fig: Response to high calcium concentration

Factors Regulating the Calcium Ion Concentration

Three factors regulate the calcium ion concentration:

- Parathyroid hormone (PTH)
- Active vitamin D (1,25-dihydroxycholecalciferol or $1,25[\text{OH}]_2\text{D}_3$)
- Calcitonin

All these three chemicals act at bone, intestine, and kidney to regulate extracellular Ca^{2+} concentration.

(a) Parathyroid hormone increases Ca^{2+} in the blood

In mammals, PTH is secreted by chief cells of parathyroid glands. PTH is an 84-amino acid peptide hormone. The chief cells produce PTH continuously and secrete it by exocytosis. Their rate of secretion depends on the concentration of Ca^{2+} in the extracellular fluid. Its concentration is increasing when extracellular Ca^{2+} is low. It is decreasing when it is high. The chief cells possess a G protein-coupled receptor on their cell membrane. These are Ca-sensing receptor (CaSR). It acts via the amplifier enzyme phospholipase C to inhibit the secretion of PTH. Thus, less PTH is secreted when

extracellular Ca^{2+} is high. When extracellular Ca^{2+} is low, the inhibition is lifted and more PTH is secreted. PTH exerts its effect at all targets by binding to cell-surface membrane receptors. It activates the amplifier enzyme **adenylyl cyclase**, and increases the second messenger cAMP.

1. In bone, PTH acts synergistically with active vitamin D to cause bone resorption (breakdown). This process releases both Ca^{2+} and phosphate into extracellular fluid.
2. In the kidney, PTH stimulates certain cells of nephron to reabsorb Ca^{2+} , and other cells to excrete phosphate anions. This combined effect ensures that reabsorbed Ca^{2+} does not complex with phosphate. Instead, it remains free and biologically active.
3. Finally, PTH stimulates certain cells of nephron to convert inactive vitamin D to its active form. One action of active vitamin D is to promote Ca^{2+} absorption from the intestine.

(a) Active vitamin D increases Ca^{2+} and phosphate in the blood.

The inactive form of vitamin D is produced in skin cells by the action of UV light. It is also obtained from the diet and from vitamin supplements. Both low extracellular Ca^{2+} and PTH stimulate conversion of inactive vitamin to its active form ($1,25(\text{OH})_2\text{D}_3$). This steroid hormone targets three main types of cells. These cells are epithelial cells of the small intestine, osteoblasts of bone, and cells in the kidney nephron.

1. Active vitamin D exerts its action through gene transcription. It synthesizes intracellular calcium-binding protein, **calbindin**. Calbindin facilitates absorption of Ca^{2+} from intestinal lumen into blood. Ca^{2+} enters blood and bind to calbindin.
2. Another action of $1,25(\text{OH})_2\text{D}_3$ is to act synergistically with PTH to release Ca^{2+} and phosphate from bone. Both hormones target bone-forming cells called **osteoblasts**. Both of these release cytokines from osteoblasts. The cytokines activates **osteoclasts**. Active osteoclasts break down bone (reabsorb it). It liberates Ca^{2+} and phosphate ions.
3. Finally, $1,25(\text{OH})_2\text{D}_3$ stimulates reabsorption of both Ca^{2+} and phosphate ions at the kidney nephron.

(c) Calcitonin opposes decreases Ca^{2+} and phosphate in the blood

Calcitonin is a 32-amino acid peptide hormone. It is secreted by C (clear) cells in the **thyroid gland** in mammals. C cells possess a G protein-coupled membrane calcium-sensing receptor similar. The high extracellular Ca^{2+} inhibits the release of PTH. It stimulates the release of calcitonin.

1. Calcitonin acts directly at the osteoclasts to inhibit bone resorption and release of Ca^{2+} and phosphate ions.
2. At the kidney nephron, calcitonin facilitates excretion of both Ca^{2+} and phosphate.

PINEAL GLAND

Its shape is like a pine cone. Therefore, it is named as pineal gland. Its cells have evolved from the photoreceptors of lower vertebrates. These cells synthesize **melatonin**. They are most active in the darkness. Light inhibits the enzyme needed for melatonin synthesis. Melatonin has a cyclical production. Therefore, melatonin can affect many physiological

the anterior pituitary. TRH directly affects the anterior pituitary gland cells to increase their output of TSH.

(c) **Effects of Cold and Other Neurogenic Stimuli on TRH and TSH Secretion**
Cold is one of the best-known stimuli for increasing the rate of TRH secretion by the hypothalamus. This effect almost certainly results from excitation of the hypothalamic centers for body temperature control. Various emotional reactions can also affect the output of TRH and TSH. Therefore, these indirectly affect the secretion of thyroid hormones.

(d) **Feedback Effect of Thyroid Hormone**
Increased thyroid hormone in the body fluids decreases secretion of TSH by anterior pituitary. When rate of thyroid hormone secretion rises to about 1.75 times normal, the rate of TSH secretion falls to zero. The increased thyroid hormone inhibits anterior pituitary secretion of TSH mainly by a direct effect on the anterior pituitary gland itself. Thus feedback mechanism maintains an almost constant concentration of free thyroid hormones in the circulating body fluids.

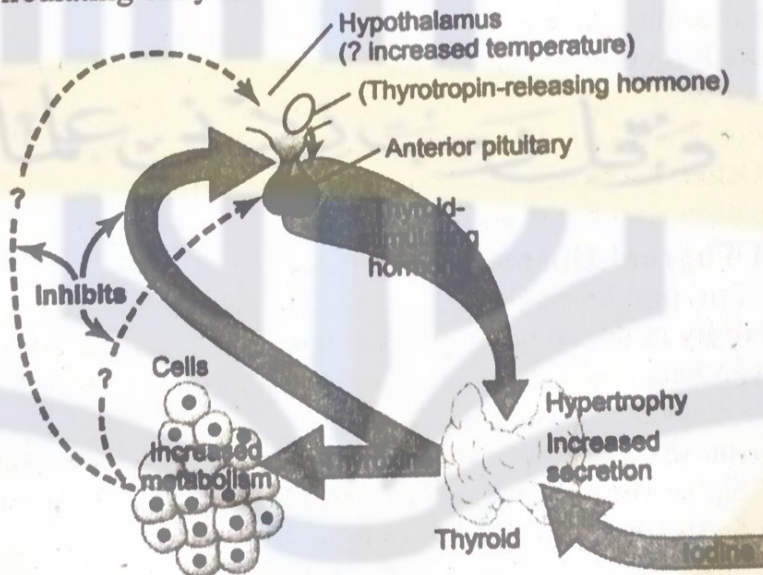


Fig: Regulation of thyroid secretion

PARATHYROID GLANDS

The parathyroid glands are tiny pea-sized glands. They are embedded in the thyroid lobes. Two glands are present in each lobe. Parathyroid glands secrete parathormone (PTH). Parathormone regulates the concentrations of calcium (Ca^{2+}) and phosphate (HPO_3^{-4}) ions in the blood.

Sometimes, there is low concentration of the calcium in the blood. The blood passes through the parathyroid glands. The parathyroid gland increases the secretion of PTH. The PTH has following effects:

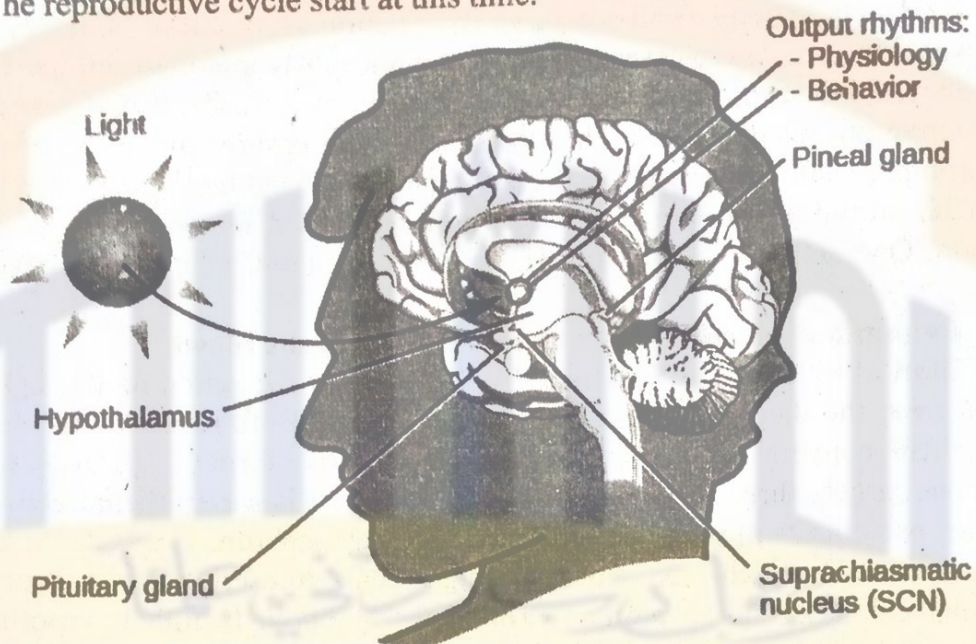
1. It stimulates the breakdown of bone cells. These bones cells release calcium ions into the blood.

activity.

Role in human

It is believed that pineal gland does play at least some role in controlling sexual drive and reproduction in humans.

processes. It adjusts the animals to diurnal and seasonal cycles. The use of melatonin is an evolutionary adaptation in mammals. It controls the periodic activities of mammals according to environment. The decreased melatonin secretion in humans starts the puberty. The reproductive cycle start at this time.



Role of Pineal Gland in Controlling Reproductive Cycles

Pineal gland is a vestigial remnant of third eye located high in the back of the head in some lower animals. Pineal gland plays a regulatory role in sexual and reproductive function. Some lower animals bear their young at certain seasons of the year. The pineal gland was removed in these animals. In such animals, normal periods of seasonal fertility were lost. Such seasonal fertility is important for these animals. It allows birth of the offspring at the time of year when survival is most likely. It has following mechanism:

- First: Pineal gland controlled by amount of light:** Pineal gland is controlled by amount of light or time pattern of light seen by the eyes each day. For example, in hamster, greater than 13 hours of darkness each day activates the pineal gland. But less than that amount of darkness fails to activate it. The nervous pathway involves the passage of light signals from eyes to suprachiasmatic nucleus of hypothalamus and then to pineal gland. It activates the pineal secretion.
- Second: pineal gland secretes melatonin and several other, similar substances.** Melatonin or other substances pass blood to anterior pituitary gland. It decreases gonadotropic hormone secretion. Thus, in the presence of pineal gland secretion, gonadotropic hormone secretion is suppressed in some species of animals. Thus the gonads become inhibited and even partly involuted. This occurs during the early winter months when there is increasing darkness. But after about 4 months of dysfunction, gonadotropic hormone secretion inhibits the effect of the pineal gland. Thus the gonads become functional once more. It is ready for a full spring-time of